1. What is the difference between / and // in Python?

/ represents precise division (result is a floating point number) whereas // represents floor division (result is an integer).

7. Can we Pass a function as an argument in Python?

Yes, Several arguments can be passed to a function, including objects, variables (of the same or distinct data types) and functions. Functions can be passed as parameters to other functions because they are objects. Higher-order functions are functions that can take other functions as arguments.

8. What is a dynamically typed language?

- In a **dynamically typed language,** the data type of a **variable is determined at runtime**, not at compile time.
- **No need to declare data types** manually; Python automatically detects it based on the assigned value.
- Examples of dynamically typed languages: **Python, JavaScript.**
- Examples of statically typed languages: **C, C++, Java.**
- **Dynamically typed languages** are easier and faster to code.
- **Statically typed languages** are usually faster to execute due to type checking at compile time.

9. What is pass in Python?

- The **pass** statement is a **placeholder that does nothing.**
- It is used when a statement is syntactically required but no code needs to run.
- Commonly used when defining empty functions, classes or loops during development.

10. How are arguments passed by value or by reference in Python?

- Python's argument-passing model is **neither "Pass by Value" nor "Pass by Reference" but it is "Pass by Object Reference".**
- Depending on the type of object you pass in the function, the function behaves differently. Immutable objects show "pass by value" whereas mutable objects show "pass by reference".

11. What is a lambda function?
A lambda function is an anonymous function. This function can have any number of parameters but, can have just one statement.
In the example, we defined a lambda function(**upper**) to convert a string to its upper case using upper().

12. What is List Comprehension?

List comprehension is a way to create lists using a concise syntax. It allows us to generate a new list by applying an **expression** to each item in an existing **iterable** (such as a **list** or **range**). This helps us to write cleaner, more readable code compared to traditional looping techniques

13. What are *args and **kwargs?

- ***args:** The special syntax **args* in function definitions is used to pass a variable number of arguments to a function. Python program to illustrate *args for a variable number of arguments:

14. What is a break, continue and pass in Python?

- **Break statement**is used to terminate the loop or statement in which it is present. After that, the control will pass to the statements that are present after the break statement, if available.
- **Continue** is also a loop control statement just like the break statement. continue statement is opposite to that of the break statement, instead of terminating the loop, it forces to execute the next iteration of the loop.
- **Pass**means performing no operation or in other words, it is a placeholder in the compound statement, where there should be a blank left and nothing has to be written there.

15. What is the difference between a Set and Dictionary?

- A Python Set is an unordered collection data type that is iterable, mutable and has no duplicate elements. Python's set class represents the mathematical notion of a set.
- **Syntax**: Defined using curly braces {} or the set() function.

my_set = {1, 2, 3}

- Dictionary in Python is an ordered (since Py 3.7) [unordered (Py 3.6 & prior)] collection of data values, used to store data values like a map, which, unlike other Data Types that hold only a single value as an element, Dictionary holds **key:value** pair. Key-value is provided in the dictionary to make it more optimized.
- **Syntax**: Defined using curly braces {} with key-value pairs.

my_dict = {"a": 1, "b": 2, "c": 3}

What are Built-in data types in Python?

The following are the standard or built-in data types in Python:

- **Numeric:** The numeric data type in Python represents the data that has a numeric value. A numeric value can be an integer, a floating number, a Boolean**,** or even a complex number.
- **Sequence Type:** The sequence Data Type in Python is the ordered collection of similar or different data types. There are several sequence types in Python:
    - Python String
    - Python List
    - Python Tuple

    - Python range

- **Mapping Types:** In Python, hashable data can be mapped to random objects using a mapping object. There is currently only one common mapping type, the dictionary and mapping objects are mutable.
    - Python Dictionary
- **Set Types:** In Python, a Set is an unordered collection of data types that is iterable, mutable and has no duplicate elements. The order of elements in a set is undefined though it may consist of various elements.

17. What is the difference between a Mutable datatype and an Immutable data type?

- Mutable data types can be edited i.e., they can change at runtime. **Eg** – List, Dictionary, etc.
- Immutable data types can not be edited i.e., they can not change at runtime. **Eg** – String, Tuple, etc.

18. What is a Variable Scope in Python?

The location where we can find a variable and also access it if required is called the scope of a variable.

- **Python Local variable:** Local variables are those that are initialized within a function and are unique to that function. A local variable cannot be accessed outside of the function.
- **Python Global variables:** Global variables are the ones that are defined and declared outside any function and are not specified to any function.
- **Module-level scope:** It refers to the global objects of the current module accessible in the program.
- **Outermost scope:** It refers to any built-in names that the program can call. The name referenced is located last among the objects in this scope.

19. How is a dictionary different from a list?

A list is an ordered collection of items accessed by their index, while a dictionary is an unordered collection of key-value pairs accessed using unique keys. Lists are ideal for sequential data, whereas dictionaries are better for associative data. For example, a list can store [10, 20, 30], whereas a dictionary can store {"a": 10, "b": 20, "c": 30}.

20. What is docstring in Python?

Python documentation strings (or docstrings) provide a convenient way of associating documentation with Python modules, functions, classes and methods.

- **Declaring Docstrings:** The docstrings are declared using "'triple single quotes"' or """triple double quotes""" just below the class, method, or function declaration. All functions should have a docstring.
- **Accessing Docstrings:** The docstrings can be accessed using the __doc__ method of the object or using the help function.

21. How is Exceptional handling done in Python?

There are 3 main keywords i.e. try, except and finally which are used to catch exceptions:

- **try**: A block of code that is monitored for errors.
- **except**: Executes when an error occurs in the try block.
- **finally**: Executes after the try and except blocks, regardless of whether an error occurred. It's used for cleanup tasks.

22. What is the difference between Python Arrays and Lists?

- Arrays (when talking about the array module in Python) are specifically used to store a collection of numeric elements that are all of the same type. This makes them more efficient for storing large amounts of data and performing numerical computations where the type consistency is maintained.
- Lists are more flexible than arrays in that they can hold elements of different types (integers, strings, objects, etc.). They come built-in with Python and do not require importing any additional modules.
- Lists support a variety of operations that can modify the list.

23. What are Modules and Packages in Python?

A module is a single file that contains Python code (functions, variables, classes) which can be reused in other programs. You can think of it as a code library. For example: **math** is a built-in module that provides math functions like sqrt(), pi, etc.

24. What is the difference between xrange and range functions?

range() and xrange() are two functions that could be used to iterate a certain number of times in for loops in Python.

- In Python 3, there is no xrange, but the range function behaves like xrange.
- In Python 2
    - **range()** – This returns a range object, which is an immutable sequence type that generates the numbers on demand.
    - **xrange()** – This function returns the generator object that can be used to display numbers only by looping. The only particular range is displayed on demand and hence called lazy evaluation.

27. Differentiate between List and Tuple?

Let's analyze the differences between List and Tuple:

List

- Lists are Mutable datatype.

- Lists consume more memory
- The list is better for performing operations, such as insertion and deletion.
- The implication of iterations is Time-consuming

Tuple

- Tuples are Immutable datatype.
- Tuple consumes less memory as compared to the list
- A Tuple data type is appropriate for accessing the elements
- The implication of iterations is comparatively Faster

28. What is the difference between a shallow copy and a deep copy?

Below is the tabular Difference between the Shallow Copy and Deep Copy:

Shallow Copy	Deep Copy
Shallow Copy stores the references of objects to the original memory address.	Deep copy stores copies of the object's value.
Shallow Copy reflects changes made to the new/copied object in the original object.	Deep copy doesn't reflect changes made to the new/copied object in the original object.
Shallow Copy stores the copy of the original object and points the references to the objects.	Deep copy stores the copy of the original object and recursively copies the objects as well.
A shallow copy is faster.	Deep copy is comparatively slower.

29. Which sorting technique is used by sort() and sorted() functions of python?

Python uses the **Tim Sort** algorithm for sorting. It's a stable sorting whose worst case is O(N log N). It's a hybrid sorting algorithm, derived from merge sort and insertion sort, designed to perform well on many kinds of real-world data.

30. What are Decorators?

Decorators are a powerful and flexible way to modify or extend the behavior of functions or methods, without changing their actual code. A decorator is essentially a function that takes another function as an argument and returns a new function with enhanced functionality.

Decorators are often used in scenarios such as logging, authentication and memorization, allowing us to add additional functionality to existing functions or methods in a clean, reusable way.

31. How do you debug a Python program?

1. Using pdb (Python Debugger):

pdb is a built-in module that allows you to set breakpoints and step through the code line by line. You can start the debugger by adding import pdb; pdb.set_trace() in your code where you want to begin debugging.

2. Using logging Module:

For more advanced debugging, the logging module provides a flexible way to log messages with different severity levels (INFO, DEBUG, WARNING, ERROR, CRITICAL).

32. What are Iterators in Python?

In Python, iterators are used to iterate a group of elements, containers like a list. Iterators are collections of items and they can be a list, tuples, or a dictionary. Python iterator implements __itr__ and the next() method to iterate the stored elements. We generally use loops to iterate over the collections (list, tuple) in Python.

33. What are Generators in Python?

In Python, the generator is a way that specifies how to implement iterators. It is a normal function except that it yields expression in the function. It does not implement __itr__ and __next__ method and reduces other overheads as well.

If a function contains at least a yield statement, it becomes a generator. The yield keyword pauses the current execution by saving its states and then resumes from the same when required.

35. What is Polymorphism in Python?

Polymorphism means the ability to take multiple forms. Polymorphism allows different classes to be treated as if they are instances of the same class through a common interface. This means that a method in a parent class can be overridden by a method with the same name in a child class, but the child class can provide its own specific implementation. This allows the same method to operate differently depending on the object that invokes it. Polymorphism is about overriding, not overloading; it enables methods to operate on objects of different classes, which can have their own attributes and methods, providing flexibility and reusability in the code.

36. Define encapsulation in Python?

Encapsulation is the process of hiding the internal state of an object and requiring all interactions to be performed through an object's methods. This approach:

- Provides better control over data.
- Prevents accidental modification of data.
- Promotes modular programming.

Python achieves encapsulation through **public**, **protected** and **private** attributes.

37. How do you do data abstraction in Python?

Data Abstraction is providing only the required details and hides the implementation from the world. The focus is on exposing only the essential features and hiding the complex implementation behind an interface. It can be achieved in Python by using interfaces and abstract classes.

38. How is memory management done in Python?

Python uses its private heap space to manage the memory. Basically, all the objects and data structures are stored in the private heap space. Even the programmer can not access this private space as the interpreter takes care of this space. Python also has an inbuilt garbage collector, which recycles all the unused memory and frees the memory and makes it available to the heap space.

39. How to delete a file using Python?

We can delete a file using Python by following approaches:

1. Python Delete File using os. remove
2. Delete file in Python using the send2trash module
3. Python Delete File using os.rmdir

40. What is slicing in Python?

Python Slicing is a string operation for extracting a part of the string, or some part of a list. With this operator, one can specify where to start the slicing, where to end and specify the step. List slicing returns a new list from the existing list.

42. What is PIP?

PIP is an acronym for Python Installer Package which provides a seamless interface to install various Python modules. It is a command-line tool that can search for packages over the internet and install them without any user interaction.

43. What is a zip function?

Python zip() function returns a zip object, which maps a similar index of multiple containers. It takes an iterable, converts it into an iterator and aggregates the elements based on iterables passed. It returns an iterator of tuples.